

# Zhuoyan Sun

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## Education

### University College London

Sep 2025 - Present

*MSc. Health Data Science (1st Degree)*

*London | UK*

- **Core Modules:** Data Science and Statistics, Artificial Intelligence in Healthcare, Advanced Machine Learning for Healthcare, Advanced Data Science and Statistics, Artificial Intelligence in Healthcare Group Project

### Imperial College London

Sep 2021 - Jun 2024

*BSc. Biological Sciences*

*London | UK*

- **Core Modules:** Mathematics, Bioinformatics, Programming and Statistics, Computational Omics, Integrative System Biology
- **Final Year Project:** Structural and Dynamic Analysis of Rare Genetic Variants in C9 and their Membrane-Binding Mechanisms Revealed by Molecular Dynamics Simulations

## Work Experience

### Associate Strategy Analyst

May 2025 - Aug 2025

*Ipsos Strategy3*

*Beijing | CN*

- Analyzed large-scale healthcare and consumer datasets from 100+ expert interviews to extract quantitative trends and evidence-based market insights
- Performed data-driven market sizing and predictive modeling to estimate \$20M revenue potential across emerging health product segments
- Collaborated with cross-functional data, research, and strategy teams to translate analytical findings into business recommendations and executive dashboards

### Research Data Scientist

Sep 2024 - Dec 2024

*Westlake University*

*Hangzhou | CN*

- Applied Python and R to process and integrate multi-omics datasets exploring epigenetic changes in aging skin stem cells
- Optimized bioinformatics pipelines, reducing computation time by 60% and enhancing data reproducibility
- Presented research findings to senior scientists, translating complex technical results into actionable biological insights

## Project Experience

### Clinical NLP & LLM Pipeline for Radiology Report

Jan 2026 - Present

*University College London*

*London | UK*

- Built a NLP pipeline using Hugging Face Transformers to parse radiology reports and generate structured features for downstream ML tasks
- Fine-tuned ClinicalBERT-style models for representation learning on clinical text, enabling quantification of diagnostic difficulty and semantic understanding
- Developed a model evaluation framework to surface failure modes in vision-language systems and benchmark reliability using text-derived signals and patient data

### Causal ML Model for Cardiovascular Risk Simulation

Dec 2025 - Present

*University College London*

*London | UK*

- Built an end-to-end ML pipeline on longitudinal biobank data, engineering multi-timepoint features to predict cardiovascular risk and model behavioural dynamics
- Developed and validated both predictive and causal ML models to generate individual-level counterfactual risk estimates for lifestyle interventions
- Designed a simulation-driven webpage translating model outputs into actionable, user-facing insights for preventive healthcare

## Skills

**Programming Skills:** Python, JavaScript, R, SQL, MySQL, Linux Shell

**Data&Visualization:** pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel

**ML&AI:** Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, XGBoost

**MLOps&Tools:** Docker, Git, GitHub, Linux, Streamlit, AWS

**Language:** English, Mandarin